Lab 7: Two-Stage Least Squares and Its Diagnostics

SOCIOL 723 — Social Statistics II — Thursday, October 8

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1 Goals

1. Build the IV estimator by hand and see that it is a ratio of two reduced-form quantities.
2. Run 2SLS properly, and understand why running the two stages manually gives the wrong standard errors.
3. Read every diagnostic an IV paper should report.
4. Watch weak instruments destroy 2SLS in simulation.
5. Characterize the compliers.

```r
library(dplyr)
library(ggplot2)
library(ivreg)
library(fixest)
library(sandwich); library(lmtest)

data("SchoolingReturns", package = "ivreg")
card <- SchoolingReturns |>
  mutate(lwage = log(wage), near4 = as.integer(nearcollege4 != "none"))
c(n = nrow(card), share_near_4yr = mean(card$near4))
#>               n share_near_4yr
#>       3010.0000         0.6821
table(card$nearcollege4)
#>
#>    none  public private
#>     957    1483     570
```

2 The design

The question is the causal return to a year of schooling. Schooling is endogenous: unmeasured ability raises both schooling and wages, so OLS is biased upward; measurement error pushes the other way.

The instrument, following Card (1995), is **growing up near a four-year college**. The argument for relevance is that proximity lowers the cost of attending. The argument for exclusion is that, conditional on region and urban status, where your parents happened to live does not affect your wage except through the schooling it made cheaper.

> **Warning**
>
> That exclusion argument is contestable, and it has been contested — college towns differ from other places in labour markets, school quality, and family background. Part of today's job is learning to interrogate it rather than accept it.

3 By hand: the Wald estimator

With a binary instrument, IV is the ratio of two differences in means.

```r
rf <- mean(card$lwage[card$near4 == 1]) - mean(card$lwage[card$near4 == 0])
fs <- mean(card$education[card$near4 == 1]) - mean(card$education[card$near4 == 0])

c(reduced_form = rf, first_stage = fs, wald_ratio = rf / fs)
#> reduced_form  first_stage   wald_ratio
```

```
#>       0.1559       0.8290       0.1881
```

```
## the same numbers as regressions
c(reduced_form = coef(lm(lwage ~ near4, data = card))["near4"],
  first_stage  = coef(lm(education ~ near4, data = card))["near4"],
  iv           = coef(ivreg(lwage ~ education | near4, data = card))["education"])
#> reduced_form.near4  first_stage.near4       iv.education
#>             0.1559             0.8290             0.1881
```

> **Note**
>
> Notice the two ingredients. The reduced form is small (about 0.16 log points) and so is the first stage (about 0.83 years). Dividing one small number by another is exactly the operation that makes IV fragile — and once we add covariates, both shrink further.

4 2SLS with covariates

```
f_ols <- lwage ~ education + experience + I(experience^2) + ethnicity +
  smsa + south

m_ols <- lm(f_ols, data = card)

m_iv <- ivreg(lwage ~ education + experience + I(experience^2) + ethnicity +
                smsa + south |
                near4 + age + I(age^2) + ethnicity + smsa + south,
              data = card)

round(cbind(OLS = coef(m_ols)[1:3], IV = coef(m_iv)[1:3]), 4)
#>                OLS     IV
#> (Intercept) 4.7337 4.0657
#> education   0.0740 0.1329
#> experience  0.0836 0.0560
```

Note the instrument set: because `experience` is a function of `education` (exper $=$ age $-$ educ $-$ 6), it is endogenous too, and we instrument it with `age`. Getting this bookkeeping right matters — every endogenous regressor needs its own instrument.

```
summary(m_iv, vcov = sandwich, diagnostics = TRUE)
#>
#> Call:
#> ivreg(formula = lwage ~ education + experience + I(experience^2) +
#>     ethnicity + smsa + south | near4 + age + I(age^2) + ethnicity +
#>     smsa + south, data = card)
```

```
#>
#> Residuals:
#>     Min      1Q  Median      3Q     Max
#> -1.8240 -0.2525  0.0229  0.2635  1.3156
#>
#> Coefficients:
#>                  Estimate Std. Error t value       Pr(>|t|)
#> (Intercept)      4.065667   0.599007    6.79 0.000000000014 ***
#> education        0.132947   0.050650    2.62        0.00871 **
#> experience       0.055961   0.025869    2.16        0.03060 *
#> I(experience^2) -0.000796   0.001326   -0.60        0.54862
#> ethnicityafam   -0.103140   0.075336   -1.37        0.17108
#> smsayes          0.107985   0.049330    2.19        0.02867 *
#> southyes        -0.098175   0.028400   -3.46        0.00055 ***
#>
#> Diagnostic tests:
#>                                    df1  df2 statistic             p-value
#> Weak instruments (education)         3 3003      8.23            0.000019 ***
#> Weak instruments (experience)        3 3003   1584.70 < 0.0000000000000002 ***
#> Weak instruments (I(experience^2))   3 3003   1114.21 < 0.0000000000000002 ***
#> Wu-Hausman                           2 3001      0.88                0.42
#> Sargan                               0   NA        NA                  NA
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 0.403 on 3003 degrees of freedom
#> Multiple R-Squared: 0.176,   Adjusted R-squared: 0.175
#> Wald test:  163 on 6 and 3003 DF,  p-value: <0.0000000000000002
```

The IV estimate is *larger* than OLS. That is the standard finding with this instrument and it is worth pausing on: if ability bias were the only problem, IV should come out *smaller*. Candidate explanations are measurement error in self-reported schooling, and the fact that the LATE applies to compliers whose return may genuinely be higher.

4.1 Why not run the two stages by hand?

To make the comparison clean, use a specification with a **single** endogenous regressor: instrument `education` with `near4`, and treat `age` (rather than `experience`) as the exogenous age control so nothing else is endogenous.

```
m_iv1 <- ivreg(lwage ~ education + age + I(age^2) + ethnicity + smsa + south |
                 near4 + age + I(age^2) + ethnicity + smsa + south, data = card)
```

```r
stage1 <- lm(education ~ near4 + age + I(age^2) + ethnicity + smsa + south,
             data = card)
card$edu_hat <- fitted(stage1)
stage2 <- lm(lwage ~ edu_hat + age + I(age^2) + ethnicity + smsa + south,
             data = card)

round(cbind(
  coefficient = c(manual = coef(stage2)["edu_hat"], ivreg = coef(m_iv1)["education"]),
  std_error = c(manual = sqrt(diag(vcov(stage2)))["edu_hat"],
                ivreg = sqrt(diag(vcov(m_iv1)))["education"])
), 5)
#>                  coefficient std_error
#> manual.edu_hat       0.09361   0.04734
#> ivreg.education      0.09361   0.04971
```

The **coefficients agree** but the **standard errors do not**. The manual second stage treats $\hat{D}$ as if it were data rather than an estimate, so it understates the uncertainty. Always use `ivreg()` or `feols()`.

4.2 The same thing in `fixest`

```r
m_fe <- feols(lwage ~ age + I(age^2) + ethnicity + smsa + south |
                education ~ near4,
              data = card, vcov = "hetero")
summary(m_fe)
#> TSLS estimation - Dep. Var.: lwage
#>                   Endo.    : education
#>                   Instr.   : near4
#> Second stage: Dep. Var.: lwage
#> Observations: 3,010
#> Standard-errors: Heteroskedasticity-robust
#>                  Estimate Std. Error t value     Pr(>|t|)
#> (Intercept)      3.275667   0.699350  4.6839 0.0000029401 ***
#> fit_education    0.093607   0.049117  1.9058 0.0567710799 .
#> age              0.082440   0.070355  1.1718 0.2413858585
#> I(age^2)        -0.000733   0.001229 -0.5966 0.5508518552
#> ethnicityafam   -0.101177   0.073452 -1.3775 0.1684746903
#> smsayes          0.108067   0.049700  2.1744 0.0297549961 *
#> southyes        -0.099427   0.029831 -3.3330 0.0008694188 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> RMSE: 0.405213   Adj. R2: 0.164384
```

```
#> F-test (1st stage), education: stat = 10.5   , p = 0.001191, on 1 and 3,003 DoF.
#>                      Wu-Hausman: stat =  1.6669, p = 0.196773, on 1 and 3,002 DoF.
```

```
fitstat(m_fe, c("ivf", "ivwald", "wh"))
#> F-test (1st stage), education: stat = 10.5   , p = 0.001191, on 1 and 3,003 DoF.
#> Wald (1st stage), education  : stat = 10.2   , p = 0.001401, on 1 and 3,003 DoF, VCOV: Het
#>                      Wu-Hausman: stat =  1.6669, p = 0.196773, on 1 and 3,002 DoF.
```

5 Diagnostics

5.1 The first stage

```
fs_mod <- lm(education ~ near4 + age + I(age^2) + ethnicity + smsa + south,
             data = card)
round(coef(summary(fs_mod))["near4", ], 4)
#>   Estimate Std. Error    t value   Pr(>|t|)
#>     0.3471     0.1070     3.2441     0.0012

## robust F on the excluded instrument
car::linearHypothesis(fs_mod, "near4 = 0", vcov. = vcovHC(fs_mod, "HC1"))
#>
#> Linear hypothesis test:
#> near4 = 0
#>
#> Model 1: restricted model
#> Model 2: education ~ near4 + age + I(age^2) + ethnicity + smsa + south
#>
#> Note: Coefficient covariance matrix supplied.
#>
#>   Res.Df Df    F Pr(>F)
#> 1   3004
#> 2   3003  1 10.2 0.0014 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Growing up near a four-year college is associated with about a third of a year more schooling once covariates are included, and the robust F is right at the traditional threshold of 10.

That is **not reassuring**. Recall from lecture that the threshold of 10 is now known to be far too lax: Lee, McCrary, Moreira, and Porter (2022) show that a conventional 5% t-test on the 2SLS estimate has correct size only when the first-stage F exceeds roughly **104**. At $F \approx 10$ we are deep in the weak-instrument regime, and the conventional standard error is badly optimistic.

```r
## the Lee et al. tF adjustment factor, interpolated from their Table 3
## Lee et al. (2022, Table 3) give ADJUSTED CRITICAL VALUES, not SE multipliers.
## The implied interval-widening factor is (critical value) / 1.96.
tF_crit <- function(F) {
  Fgrid <- c(4, 6, 8, 10, 12, 15, 20, 30, 40, 60, 100, 104.7, 200)
  crit  <- c(Inf, 18.66, 5.72, 3.43, 3.02, 2.71, 2.54, 2.31, 2.20, 2.11, 2.01,
             1.96, 1.96)
  approx(Fgrid, crit, xout = F, rule = 2)$y
}
Fstat <- car::linearHypothesis(fs_mod, "near4 = 0",
                               vcov. = vcovHC(fs_mod, "HC1"))$F[2]
se_iv <- sqrt(diag(vcovHC(m_iv1, "HC1")))["education"]
b_iv  <- coef(m_iv1)["education"]
crit  <- tF_crit(Fstat)

c(F = Fstat, adjusted_crit_value = crit, widening_factor = crit / 1.96,
  estimate = b_iv, conventional_se = se_iv,
  conventional_t = b_iv / se_iv,
  significant_at_5pct = abs(b_iv / se_iv) > crit)
#>                          F          adjusted_crit_value
#>                   10.22350                      3.38418
#>            widening_factor           estimate.education
#>                    1.72662                      0.09361
#>     conventional_se.education      conventional_t.education
#>                    0.04912                      1.90580
#> significant_at_5pct.education
#>                    0.00000
```

At $F \approx 10$ the honest critical value is about 3.4 rather than 1.96 — so the interval is roughly 1.75 times wider, and a t-statistic that looked comfortably significant may not clear the bar. This is the single most important diagnostic in the lab.

Note also that `tF` is designed for the **just-identified, single-endogenous- regressor** case, which is why we built `m_iv1` above.

5.2 The reduced form

```r
rf_mod <- lm(lwage ~ near4 + age + I(age^2) + ethnicity + smsa + south,
             data = card)
round(coef(summary(rf_mod))["near4", ], 4)
#>   Estimate Std. Error    t value   Pr(>|t|)
#>     0.0325     0.0164     1.9774     0.0481
```

> **Tip**
>
> **Always report the reduced form.** It is the one regression in an IV design that requires no instrument-strength assumption at all. Here the adjusted reduced-form coefficient is about 0.032 with a standard error of 0.016 — a t of roughly 2. That is the entire evidentiary basis for the IV estimate, and it is thin. If the instrument has no detectable effect on the outcome, the IV estimate is a small number divided by another small number, and its apparent precision is an artefact.

5.3 Overidentification

```
m_over <- ivreg(lwage ~ education + experience + I(experience^2) + ethnicity +
                  smsa + south |
                  near4 + nearcollege2 + age + I(age^2) + ethnicity + smsa + south,
                data = card)
summary(m_over, diagnostics = TRUE)$diagnostics
#>                                  df1  df2 statistic    p-value
#> Weak instruments (education)       4 3002     6.090 0.00007048
#> Weak instruments (experience)      4 3002  1209.350 0.00000000
#> Weak instruments (I(experience^2)) 4 3002  1104.549 0.00000000
#> Wu-Hausman                         2 3001     1.505 0.22226630
#> Sargan                             1   NA     3.245 0.07163864
```

Read the Sargan row carefully. It also assumes homoskedasticity, unlike everything else in this lab — the heteroskedasticity-robust analogue is Hansen's J. And non-rejection is weak evidence either way: it is entirely consistent with both instruments being invalid in the same direction.

6 Weak instruments, by simulation

The cleanest way to see the danger is to build a world where you know the answer.

```
set.seed(723)

sim_iv <- function(n = 500, pi = 0.5, rho = 0.6, beta = 1) {
  z <- rnorm(n)
  u <- rnorm(n)                              # confounder
  d <- pi * z + rho * u + rnorm(n)           # pi = instrument strength
  y <- beta * d + u + rnorm(n)               # TRUE effect is 1
  m <- ivreg(y ~ d | z)
  fs <- summary(lm(d ~ z))$fstatistic[1]
  c(iv = coef(m)["d"], ols = coef(lm(y ~ d))["d"], F = fs)
}
```

```r
strengths <- c(strong = 0.5, moderate = 0.15, weak = 0.05)
res <- lapply(strengths, function(p) t(replicate(600, sim_iv(pi = p))))

sapply(res, function(r) c(
  median_F = median(r[, "F.value"]),
  mean_IV  = mean(r[, "iv.d"]),
  median_IV = median(r[, "iv.d"]),
  mean_OLS = mean(r[, "ols.d"])
)) |> round(3)
#>           strong moderate  weak
#> median_F  92.231    8.114 1.023
#> mean_IV    1.004    0.922 0.895
#> median_IV  1.013    0.974 1.206
#> mean_OLS   1.374    1.437 1.439
```

```r
bind_rows(lapply(names(res), function(nm)
  tibble::tibble(est = res[[nm]][, "iv.d"], strength = nm))) |>
  filter(abs(est) < 5) |>
  ggplot(aes(est)) +
  geom_histogram(bins = 60, fill = "navy", alpha = .7) +
  geom_vline(xintercept = 1, linetype = 2) +
  facet_wrap(~ factor(strength, levels = c("strong", "moderate", "weak")),
             scales = "free_y") +
  labs(x = expression(hat(beta)[2*SLS]), y = "count") +
  theme_minimal(base_size = 9)
```

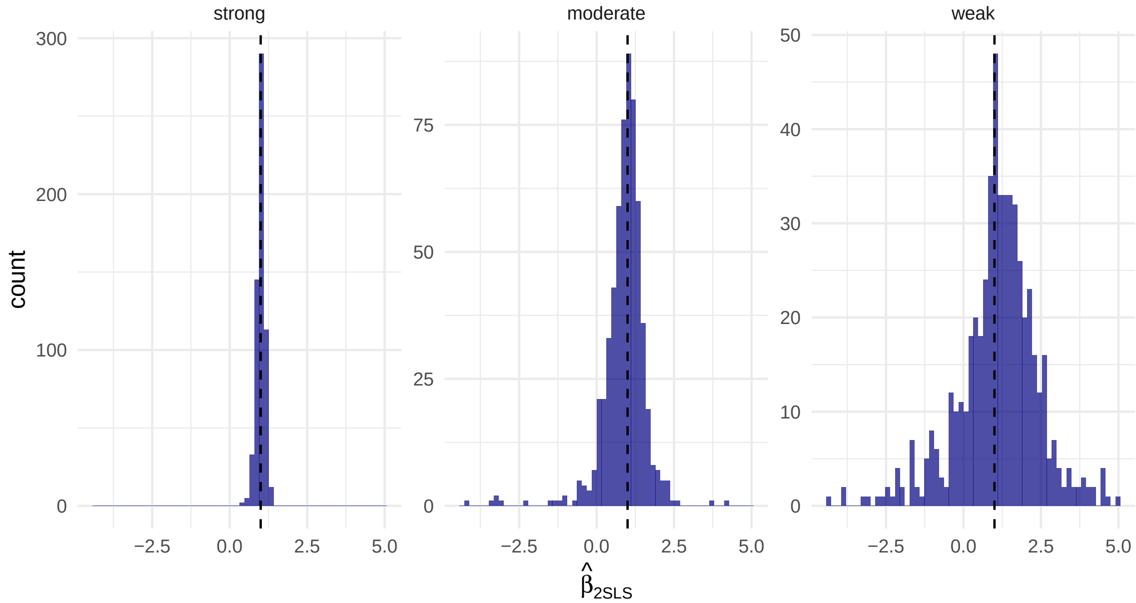

Figure 1: Sampling distribution of the 2SLS estimate at three instrument strengths, truncated to [-5, 5] for display. The truth is 1 (dashed). As the instrument weakens, 2SLS collapses toward OLS and develops very heavy tails – see the share truncated, reported below.

```
sapply(res, function(r) mean(abs(r[, "iv.d"]) >= 5))   # share outside the plot
#>   strong moderate     weak
#>   0.0000   0.0050   0.1017
```

Three lessons:

- With a **strong** instrument, 2SLS is centred on the truth and roughly normal.
- With a **weak** instrument, the distribution is badly skewed and shifted toward the OLS probability limit. IV is biased *toward the thing it was supposed to fix*.
- The tails are heavy enough that the mean is not even a useful summary. Conventional standard errors are hopeless here.

```
## coverage of nominal 95% intervals
cov_check <- function(pi) {
  out <- replicate(600, {
    z <- rnorm(500); u <- rnorm(500)
    d <- pi * z + 0.6 * u + rnorm(500); y <- 1 * d + u + rnorm(500)
    m <- ivreg(y ~ d | z)
    ci <- confint(m)["d", ]
    ci[1] < 1 & 1 < ci[2]
  })
  mean(out)
}
sapply(strengths, cov_check)
#>   strong moderate     weak
```

```
#>    0.9483    0.9683    0.9983
```

A nominal 95% interval that covers 60% of the time is not a 95% interval.

7 Who are the compliers?

The LATE applies to people whose schooling was changed by college proximity. We cannot name them, but we can describe them.

```
## complier share = first stage on a binary treatment
card2 <- card |> mutate(some_college = as.integer(education >= 13))
fs_bin <- lm(some_college ~ near4 + age + I(age^2) + ethnicity + smsa + south,
             data = card2)
round(coef(summary(fs_bin))["near4", ], 4)
#>    Estimate Std. Error    t value   Pr(>|t|)
#>      0.0599     0.0205     2.9211     0.0035
```

About 5 percent of the sample are compliers on this definition — that is, roughly 5 percent attend some college *because* they grew up near one.

```
## Abadie's kappa weights let us describe complier characteristics
pz <- fitted(glm(near4 ~ age + I(age^2) + ethnicity + smsa + south,
                 data = card2, family = binomial))
kappa <- 1 - card2$some_college * (1 - card2$near4) / (1 - pz) -
  (1 - card2$some_college) * card2$near4 / pz

profile <- function(x) c(all = mean(x), compliers = sum(kappa * x) / sum(kappa))
round(rbind(
  father_educ = profile(card2$feducation |> tidyr::replace_na(mean(card2$feducation, na.rm =
  south = profile(as.integer(card2$south == "yes")),
  smsa  = profile(as.integer(card2$smsa == "yes")),
  black = profile(as.integer(card2$ethnicity == "afam"))
), 3)
#>               all  compliers
#> father_educ 9.989    12.331
#> south       0.404     0.177
#> smsa        0.713     0.932
#> black       0.234     0.129
```

! Important

The complier profile is the interpretive heart of an IV paper. If the compliers are disproportionately from low-education families, then the LATE is a return to college **for people at the margin of attending** — which is often the policy-relevant group, but is emphatically not the ATE.

8 Exercises

1. **The bookkeeping.** Refit the IV model treating `experience` as exogenous (i.e. not instrumenting it). How much does the education coefficient move? Explain why `experience` cannot be exogenous if `education` is not.

2. **Reduced form first.** Suppose a referee reports only the 2SLS estimate and its t-statistic. Compute the reduced form and first stage and write two sentences a referee could use to evaluate whether the design is credible.

3. **A placebo.** Growing up near a college should not affect variables determined before college attendance. Regress `feducation`, `meducation`, and `library14` on `near4` with the same controls. Do any of them move? What would it mean for exclusion if they did?

4. **Anderson–Rubin.** Write a function that, for a grid of β_0 values, regresses $Y - \beta_0 D$ on the instrument and controls and records the p-value on the instrument. Plot it, and read off the 95% AR confidence set. Compare with the conventional 2SLS interval.

5. **Bias amplification with weak instruments.** In `sim_iv()`, add a small direct effect of Z on Y (violating exclusion by, say, 0.05). Show that the resulting bias in 2SLS grows as the instrument weakens, and compare it with the OLS bias.

6. **Many weak instruments.** Add 20 pure-noise instruments to the strong design. What happens to the first-stage F, and to the 2SLS estimate? Explain in terms of overfitting the first stage.

9 Session information

```r
sessionInfo()
#> R version 4.4.1 (2024-06-14)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS 26.5.2
#>
#> Matrix products: default
#> BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: Asia/Shanghai
#> tzcode source: internal
```

```
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] lmtest_0.9-40  zoo_1.8-12     sandwich_3.1-1 fixest_0.12.1  ivreg_0.6-5
#> [6] ggplot2_4.0.0  dplyr_1.1.4
#>
#> loaded via a namespace (and not attached):
#>  [1] tidyr_1.3.1         utf8_1.2.4          generics_0.1.3
#>  [4] dreamerr_1.5.0      lattice_0.22-7      digest_0.6.37
#>  [7] magrittr_2.0.3      evaluate_1.0.0      grid_4.4.1
#> [10] estimability_1.5.1  RColorBrewer_1.1-3  mvtnorm_1.3-1
#> [13] fastmap_1.2.0       Matrix_1.7-0        jsonlite_1.8.9
#> [16] Formula_1.2-5       tinytex_0.53        survival_3.6-4
#> [19] multcomp_1.4-29     purrr_1.0.2         fansi_1.0.6
#> [22] scales_1.4.0        TH.data_1.1-4       stringmagic_1.2.0
#> [25] codetools_0.2-20    numDeriv_2016.8-1.1 abind_1.4-8
#> [28] cli_3.6.5           rlang_1.3.0         splines_4.4.1
#> [31] withr_3.0.1         yaml_2.3.10         tools_4.4.1
#> [34] coda_0.19-4.1       vctrs_0.6.5         R6_2.5.1
#> [37] lifecycle_1.0.4     emmeans_1.11.1      car_3.1-3
#> [40] MASS_7.3-60.2       pkgconfig_2.0.3     pillar_1.9.0
#> [43] gtable_0.3.6        glue_1.7.0          Rcpp_1.1.1
#> [46] xfun_0.47           tibble_3.2.1        tidyselect_1.2.1
#> [49] knitr_1.48          xtable_1.8-4        farver_2.1.2
#> [52] htmltools_0.5.8.1   nlme_3.1-164        labeling_0.4.3
#> [55] rmarkdown_2.28      carData_3.0-5       compiler_4.4.1
#> [58] S7_0.2.0
```